

1.1 Introduction (Simulation)

Simulation is the process of **imitating a real-life system using a model** so that the system can be **studied, tested, and understood** without using the real system. In simple words, simulation means **behaving like the real system in a safe and low-cost way**.

Simulation is commonly used in daily life without noticing it. A **globe** represents the Earth, **scale models of machines** are used in laboratories, **model airplanes in wind tunnels** help study airflow, **children's traffic parks** imitate real traffic, **planetariums** show planetary motion, and **museums or geological parks** represent natural environments. In all cases, the **real system is replaced by a model**.

Many real systems are **too complex to model physically**, such as inventory systems, bank operations, queuing systems, manufacturing units, and managerial decision systems. Therefore, **computer simulation** is used, where the system is represented using **mathematical equations, logical rules, and computer programs**.

System simulation means **imitating the operation of a real system on a computer** and performing experiments **without disturbing the actual system**. Examples include bank operations, factory production lines, and sales force assignment. Simulation is **more flexible and practical** than purely theoretical analysis, and modern simulation software allows even **non-programmers** to analyze systems.

1.2 The System

A **system** is a **collection of interconnected components** that **interact regularly** to achieve a common objective. In simple terms, a system consists of **dependent objects working together**.

Examples of Systems

Engine Governor System:

Its purpose is to **maintain constant engine speed**. When load changes, speed varies, the governor adjusts the fuel supply, and the speed becomes stable. This is an **automatic control system**.

Manufacturing System:

It includes purchasing, production, assembly, packaging, quality control, shipping, and personnel. All departments are interconnected, so a change in one affects others, making the system **complex**.

Types of Systems

Physical Systems:

These consist of **real, tangible components** and obey physical laws. Examples include engines, governors, and fuel pumps.

Non-Physical Systems:

These consist of **abstract components** with man-made relationships, such as management systems, education systems, social systems, and political systems.

Systems Approach

Since each part of a system affects other parts, the system must be studied **as a whole rather than in isolation**. This method is known as the **systems approach** or **scientific approach**.

System and Environment

The **system** is the part under study, while the **environment** includes everything outside the system that influences it, such as raw material supply, power availability, and market demand.

Open and Closed Systems

- **Closed system:** Not affected by the environment
- **Open system:** Influenced by the environment (most real systems)

System Boundary

A **system boundary** separates the system from its environment. The boundary depends on **the purpose of the study**.

Endogenous and Exogenous Activities

- **Endogenous activities:** Occur inside the system
- **Exogenous activities:** Originate outside the system

Important System Terminology

Entity: The object of interest in a system (e.g., vehicles in traffic).

Attribute: A property of an entity such as speed, size, type, or distance.

Activity: An action that changes an attribute, such as driving, processing, or serving customers.

State of System: The condition of the system at a particular time, e.g., number of customers waiting or machines busy.

Queue: A waiting line, such as vehicles at a petrol pump or customers in a bank.

Creating: The arrival of a new entity into the system, e.g., a customer entering a bank.

Scheduling: Assigning events to entities, such as service start and completion times.

Random Variable: A variable whose value is uncertain, such as customer arrival time or bulb life.

Random Variate: A value generated from a random variable using random numbers and probability rules.

Distribution: A mathematical function describing randomness, commonly the **exponential distribution** in queuing systems.

Petrol Pump System (Example)

- **Entities:** Vehicles
- **Attributes:** Vehicle type, fuel type
- **Events:** Arrival, service start, service completion, departure
- **States:** Number of vehicles waiting, workers busy or idle
- **Queue:** Vehicles waiting for service
- **Random variables:** Arrival time, service time
- **Distribution:** Usually exponential

Table 1.1: System Examples

System	Entities	Events	State
Bank	Customers	Arrival, departure	Customers waiting
Traffic	Vehicles	Stop, go	Signal status
Production	Machines	Start, finish	Machine busy
Supermarket	Customers	Checkout	Stock level
Communication	Messages	Send, receive	Messages waiting

1.3 Continuous and Discrete Systems (Compact Exam Notes)

From the simulation point of view, systems are mainly classified into **two types: continuous systems and discrete systems**.

1. Continuous System

In a continuous system, the **state of the system changes smoothly and continuously with time**. Changes do not occur suddenly, and variables can take **any value** within a range. Most **engineering and physical science systems** fall into this category. In such systems, variables are generally **deterministic**, and the system behavior is usually represented using **mathematical equations**.

Examples:

- **Pure pursuit problem:** The positions of the target and pursuer change continuously with time.

2. Discrete System

In a discrete system, the **state changes suddenly at specific points in time**. Changes occur in steps, and events happen at **separate moments** rather than continuously. These systems often involve

countable quantities and are common in **operations research and management science**. Discrete systems are usually **stochastic**, meaning they involve randomness or uncertainty.

Examples:

- **Inventory system:**
 - Demand occurs at fixed times
 - Stock is replenished at fixed times
 - Items change in countable numbers
- **Queuing system:** Customers arrive and leave at specific times

Important Observations

Most real-life systems are **neither purely continuous nor purely discrete**.

For example, in a **reservoir system**, sudden rainfall or sudden water release causes abrupt changes, requiring discrete treatment. In a **factory system**, machine start and stop operations are discrete, while the machining process itself is continuous. Therefore, **no fixed rule exists** for modeling systems. A balance must be maintained between **model simplicity** and **accuracy of results**.

1.4 Types of Models

Models used for system analysis are classified in different ways. The main classification is into **physical models** and **mathematical models**. Each type of model can further be **static** (no change with time) or **dynamic** (changes with time).

1.4.1 Physical and Mathematical Models

1. Physical Models

Physical models use **real physical objects** to represent a system. These models are based on **similarity or analogy** with the actual system and may or may not have moving parts. A physical structure is constructed to study the system behavior.

Examples:

- Mechanical system represented by an electrical system
- Hydraulic system represented by an electrical system

Analogy example:

- Pressure → Voltage
- Flow → Current
- Resistance → Load

2. Mathematical Models

Mathematical models represent a system using **symbols, variables, and mathematical equations**. System properties are described by **independent and dependent variables**, and their relationships are expressed using **mathematical functions**. Depending on how solutions are obtained, mathematical models may be **static or dynamic, analytical, numerical, or simulation-based**.

1.4.2 Static and Dynamic Models

Static Models

A **static model** represents a system that **does not change with time**. It describes the system at **one specific moment**, and **time is not included** in the analysis.

Static models are usually based on **mathematical equations**, require **only one calculation**, and use **average values** of system variables. System performance is obtained by **adding individual effects** of different components.

Limitations:

Static models cannot analyze **time-based changes**, cannot show the effect of **parameter variation**, and ignore **interaction or synergy** among system components, where combined effects may differ from simple addition.

Use: Static models are suitable when **time is not important**.

Dynamic Models

A **dynamic model** represents a system whose behavior **changes with time**. It shows how system variables evolve and is generally implemented using **computer software**.

Dynamic models repeatedly **recalculate equations over time**, are **interactive**, allow testing of **different decisions**, and can include **randomness and uncertainty**. They help predict system performance and are important because most real-life systems behave **randomly**.

Difference between Static and Dynamic Models

Static Model	Dynamic Model
Time independent	Time dependent
One-time calculation	Repeated calculation
Uses averages	Uses changing values
Simple	More realistic

1.4.3 Analytical and Numerical Models

This classification depends on **how the mathematical model is solved**.

Analytical Models

Analytical models are solved using **mathematical formulas**, providing **exact solutions**. In some cases, a computer is not required. Examples include problems solved using **algebra or calculus**.

Numerical Models

Numerical models cannot be solved exactly using formulas and require **numerical methods** and **computer calculations**. These methods use **step-by-step computations** and are closely related to **simulation**.

Simulation is mostly applied to **mathematical models**, especially **dynamic and stochastic systems**.

1.4.4 Deterministic and Stochastic Models

Deterministic Models

In deterministic models, all inputs are **known and fixed**, and the output is **certain**. The same input always produces the **same output**, and no randomness is involved.

Stochastic Models

Stochastic models contain **one or more random inputs**, so outputs are also **random**. The results are **estimates**, not exact values.

Examples include **queuing systems**, where arrival times are random, and **production systems**, where operation times vary. Random variables may be **continuous or discrete**, and simulations must run for a **long time** to obtain reliable results.

1.5 System Simulation

Simulation

Simulation is the process of creating a **model of a real system** and studying its behavior to obtain results. It can be carried out using **physical models**, **mathematical models**, or **numerical methods**.

System Simulation

System simulation focuses on studying **system behavior over time** and mainly uses **dynamic models**. **Definition (Geoffrey Gordon):**

System simulation is the technique of solving problems by observing the performance of a **dynamic model over time**.

The two essential elements of system simulation are:

1. **Dynamic model**
2. **Time**

Important Note

In some cases, **time may not be important**, but the system is still analyzed **step-by-step**. For example, determining the required number of repair workers for a bus fleet involves sequential evaluation of different crew sizes.

Computer Simulation

System simulation is generally called **computer simulation**, as practical simulation work is performed using computers.

1.6 Computer Simulations

Many real problems can be solved using **mathematical models**. When these models can be solved exactly using formulas, they are called **analytical methods**. Even if a computer is used to solve such equations, it is **not called computer simulation** as long as the solution is exact or closed-form.

Computer simulation is used when an **exact analytical solution is not possible** or when the system is **too complex** to be handled mathematically. In such cases, simulation works either as a **replacement** for analytical models or as a **supporting tool**.

Computer simulation is widely applied in **manufacturing, business, defence, healthcare, and management**. Many of these applications involve **discrete-event simulation**, where changes occur at specific points in time.

Simulation becomes necessary because complex systems cannot be described fully using equations. It allows **IF and WHAT-IF analysis**, such as studying the effect of parameter changes. It is commonly used in **bottleneck analysis, theory of constraints, and management consulting**.

Many **software packages** are available for simulation. Most computer simulations are **human-out-of-loop simulations**, where no human interaction occurs during execution. Simulation is also widely used in **economics, social sciences, and engineering**.

A special application is **simulating computers using computers**, called **computer architecture simulation**, which includes **instruction-set simulators** and **full-system simulators**. Another modern term is **synthetic environment**, referring to **virtual or computer-generated environments**.

1.7 Physical and Interactive Simulation

Physical simulation uses **physical objects instead of real systems** because they are **smaller, cheaper, and safer**.

Example: **Testing an airplane wing in a wind tunnel**.

Interactive simulation is a special type of physical simulation where a **human actively interacts** with the model. The human is a **necessary part of the system**, so it is also called **human-in-the-loop simulation**.

Examples include **flight simulators** and **driving simulators**.

1.8 Real-Time Simulation

In many cases, complete simulation of a system is **too complex or impractical**. In such situations, **real physical components are combined with computer simulation**.

In real-time simulation, the computer simulates only the **missing parts** of the system, while humans interact with it. The simulation runs in **real time**, meaning the time taken matches actual system time.

Examples include **aircraft cockpit simulators** and **zero-gravity training systems**. These simulations require **real-time computers**, which must respond immediately and send signals at **precisely correct times**.

1.9 Simulation vs Analytical Methods

Simulation models are **run like experiments**, not solved mathematically. They provide **specific and approximate results** for given inputs and usually require **multiple runs** to improve accuracy.

Analytical models are **solved using mathematical equations** and give **general and exact solutions**. Often, a single solution provides the **optimum result**.

Comparison

Simulation	Analytical
Experiment-like execution	Mathematical solution
Approximate results	Exact results
Needs repeated runs	Single solution
Handles complex systems	Limited by assumptions

For complex systems, analytical methods require many assumptions, which may lead to **unrealistic results**. Simulation is more effective in such cases.

Simulation accuracy depends on **model detail**. More detail increases **complexity and computation time**, so a balance between **detail, accuracy, and time** is required.

Key idea: When all other methods fail, **simulation can still be used**.

1.10 When to Use Simulation

Simulation should be used:

1. To study **internal interactions** of complex systems
2. To test **new designs or policies** before real implementation
3. To **verify analytical solutions**

4. To observe effects of **input changes on outputs**
5. To suggest **system improvements** for better performance

1.11 Steps in Simulation Study

Simulation must be carried out in a **systematic, step-by-step manner**. A typical simulation study consists of **nine main steps**, followed in sequence to ensure correctness and usefulness of results.

Main Steps

1. Problem formulation
2. Model construction
3. Data collection
4. Model programming
5. Validation
6. Design of experiment
7. Simulation run and analysis
8. Documentation
9. Implementation

1.11.1 Problem Formulation

Problem formulation involves **clearly defining the problem**, setting the **objectives**, identifying **possible alternatives**, and deciding **how results will be evaluated**. The analyst and policy makers must have a **clear and common understanding** of the problem. At this stage, it must be decided whether **simulation is appropriate**, along with the selection of **performance measures, analysis methods, and expected outcomes**. A **well-defined problem** is essential for a **successful simulation study**.

1.11.2 Model Construction

Model construction is considered more an **art than a science**, and there are **no fixed rules** for building models. The key task is to identify the **important features** of the system, make suitable **assumptions and simplifications**, and begin with a **simple model**, which is then improved gradually.

Building Block Method

- Divide the system into **small sub-models**
- Validate each block individually

- Combine blocks to form the complete model

1.11.3 Data Collection

Data is a **critical requirement** for simulation. It may be obtained from **existing records** or through **new data collection**. Data requirements often **change as the model evolves**, and data is used both for **model input** and **model validation**. Since data collection is **time-consuming**, it should **begin at an early stage** of the study.

1.11.4 Model Programming

Simulation models involve **large computations and memory requirements**, making the use of **computers essential**. Model programming refers to converting the conceptual model into a **computer-executable form**.

Tools Used

- Simulation languages
- Special-purpose simulation software
- General-purpose languages:
 - BASIC
 - FORTRAN
 - C / C++

The choice of tool depends on the **problem nature**, **availability of software**, and **required flexibility**.

1.11.5 Validation

Validation ensures that the **model accurately represents the real system** and that the **program logic is correct**. It is performed by comparing simulation results with **actual system data**, identifying errors, correcting them, and repeating the process.

Validation is an **iterative process**, involving continuous **testing and debugging**, and relies heavily on **real-world data**.

1.11.6 Design of Experiment

Simulation is equivalent to conducting **experiments on a model**, and the outputs are usually **stochastic (random)**. To obtain reliable results, careful planning of the experiment is required.

Decisions Involved

- Length of simulation run

- Initial conditions
- Number of runs
- Number of replications
- Removal of initial bias
- Use of variance reduction techniques

The objective is to achieve results with **acceptable error** and a **high confidence level**.

1.11.7 Simulation Run and Analysis

In this step, the simulation program is executed, output data is collected, and results are analyzed. Based on the analysis, decisions are made regarding the need for **additional runs** or **modifications in experimental design**. This step also contributes to validating the **experiment setup**.

1.11.8 Documentation

Documentation is a **crucial part** of simulation studies. It supports **future use, reuse, and modification** of the model. Proper documentation explains the **model structure, inputs, and outputs**, and enables users to **change parameters** and study their effects. It should be clear enough for **new users** to understand easily.

1.11.9 Implementation

Implementation involves applying simulation results to the **real system**. This step becomes easier when users clearly understand the **model and assumptions**. Best practice suggests involving users **from the beginning** of the simulation study to ensure smooth implementation.

1.12 Phases of a Simulation Study

The nine steps of simulation can be grouped into **four main phases**.

Phase 1: Problem Formulation

- Problem definition

Phase 2: Model Building

- Model construction
- Data collection
- Programming
- Validation

Phase 3: Running the Model

- Design of experiment
- Simulation run
- Result analysis

Phase 4: Implementation

- Documentation
- Implementation

1.13 Advantages of Simulation

1. Allows study of a real system **without using the actual system**
 - Example: Aircraft testing in wind tunnel
2. Many managerial systems are **too complex for mathematical analysis**
3. Real-life experimentation may be **impossible, costly, or dangerous**
4. Simulation can:
 - **Compress time** (years into minutes)
 - **Expand time** (very fast processes made observable)
5. Helps identify **bottlenecks and future problems** before implementation
6. Requires **less mathematics**, making it easier to understand
7. Simulation models are **flexible**
8. Easier to use than complex mathematical models
9. Availability of many **ready-made simulation software tools**
10. Acts as an excellent **training tool**
 - Pilot training
 - Astronaut training
 - Operation of complex systems

1.14 Limitations of the Simulation Technique

Although simulation has many advantages, it also has several **important limitations**. Due to these limitations, simulation is often used as a **last option**, only when analytical or mathematical methods are not suitable.

Main Limitations

1. **No optimum solution**

Simulation does not provide the best or optimum solution. It only produces **approximate results**, and these results contain **statistical errors**, especially when uncertainty or randomness is involved.

2. **Difficulty in quantifying variables**

Certain factors such as **human behavior, motivation, and judgment** cannot be easily measured or expressed numerically, making simulation difficult.

3. **Very complex for large systems**

Large systems involve many variables and inter-relationships, which makes the simulation model **highly complex and difficult to manage**.

4. **Costly and time-consuming**

Simulation requires **long computer time, skilled manpower**, and high development effort, making it more expensive than analytical techniques.

5. **Over-dependence on simulation**

Sometimes simulation is used even when **simple mathematical methods are sufficient**, which makes the use of simulation **uneconomical**.

Exam line: Simulation gives **estimates**, not exact or optimum answers.

1.15 Areas of Application of Simulation

Simulation is applied in **almost every field**, especially where systems are complex, dynamic, and uncertain.

Major Application Areas

Manufacturing

- Production system design
- Inventory and materials management
- Capacity and layout planning
- Quality evaluation

Business

- Market analysis
- Consumer behavior prediction
- Marketing strategy optimization
- Logistics planning

Military

- War strategy testing
- Air, sea, and land operations
- Inventory management
- Simulated war exercises

Healthcare

- Hospital planning
- Patient flow analysis
- Staffing decisions
- Health program evaluation

Communication

- Network design and optimization
- Reliability analysis
- Buffer sizing
- Manpower planning

Computer Systems

- Hardware design
- Operating system design
- Networking and sharing

Economics

- Portfolio management
- Government policy analysis

- Budgeting
- Market forecasting

Transportation

- Road, rail, and air network design
- Traffic planning
- Timetable evaluation

Environmental Systems

- Waste management
- Pollution control
- Environmental program evaluation

Biological Systems

- Population studies
- Spread of diseases

Exam note: Simulation should be used **after checking mathematical methods**, since it is **costly and approximate**.

1.16 Simulation — A Management Laboratory

Managerial problems are usually **complex, random, and inter-related**, and analytical methods require **many assumptions and heavy simplifications**. In many cases, simulation becomes the **only practical tool** for analysis.

Simulation is called a **management laboratory** because managers can **test different strategies, compare alternatives, and observe results** without disturbing the real system or taking risks.

Management Games

- Used for management training
- Widely used in universities and management institutes

 Simulation allows **safe experimentation**, just like a laboratory.

1.17 Simulation in Design

Simulation is widely used in **system and product design** to evaluate designs **before actual manufacturing**, thereby reducing cost and errors.

Applications in Design

- Digital ICs are simulated before fabrication
- Robots and automated systems are tested through simulation
- Manufacturing systems are validated using simulation
- Assembly line balancing is performed using simulation
- Civil engineering structures are tested before construction

Key Advantage

Simulation helps detect **design errors at an early stage**. Early correction is **cheap**, whereas errors found late in the design process are **very expensive**.

Exam point: Simulation should be used **early in the design cycle**.

1.18 Simulation in Computer Science

In computer science, simulation refers to **executing a program step-by-step** and observing its **inputs, outputs, and state changes**.

Uses in Computer Science

- Computer architecture design
- Testing different system designs
- Performance optimization

Computer Architecture Simulation

Simulation is used when real machines are **expensive or inconvenient**. Simulators help in **testing programs, debugging software, and designing processors**.

VLSI and Circuit Design

All VLSI circuits are **simulated and tested** before actual manufacturing to ensure correct operation.

Fault Analysis

Simulation supports **fault tree analysis** and helps improve **system reliability**.

1.19 Simulation in Training

Simulation is widely used for **training purposes**, especially when real-life training is **dangerous** or **very expensive**. Instead of exposing trainees to real risks, simulation provides a **safe and controlled environment**.

Examples

- Pilots are trained first using **flight simulators**
- Astronauts practice in **zero-gravity virtual laboratories**
- Military forces use **simulated combat environments** instead of real war

Purpose of Training Simulation

- Safely replicate **real-life situations**
- Test and improve **decision-making ability**
- Develop fast and correct **reaction under pressure**

1.20 Classification of Training Simulations

Training simulations are broadly classified into **three main types**.

1. Live Simulation

- Uses **real people, real equipment**, and the **real world**
- Example: Soldiers performing mock drills using actual weapons
- Main focus: **Operating real equipment in real conditions**

2. Virtual Simulation

- Uses **real people**, but **simulated equipment and environment**
- Examples:
 - Astronauts training with mock spacecraft in virtual space
 - Military virtual war games
- Advantages:
 - Safer than live simulation
 - Cheaper
 - Allows risky actions without real damage

- Main focus: **Practice in a virtual environment**

3. Constructive Simulation

- Uses **simulated people, simulated equipment, and simulated environment**
- Examples:
 - Table-top war games
 - Command and staff exercises
- Main focus:
 - **Command, planning, and decision-making**
 - Training commanders to handle unpredictable and fully equipped enemies

Important exam line: Live & Virtual simulations train **individuals**, while Constructive simulations train **commanders and decision makers**.

1.21 Simulation in Education

Simulation in education is an **interactive learning tool** that helps students **understand concepts more clearly** than traditional methods.

A famous saying explains its importance:

"Tell me, I forget. Show me, I remember. Involve me, I understand."

Advantages

1. Interactive and flexible teaching method
2. Students learn at **their own pace**
3. Safe practice through **virtual labs and simulations**
4. Experiments can be repeated **many times without risk**

Blended Education

Blended education combines:

- Classroom teaching
- Internet-based learning
- Simulation-based modules

The goal is to achieve **maximum learning effectiveness** by combining the strengths of all methods.

Examples

- Virtual laboratory experiments
- Animated tools for engineering drawings
- Educational video games for school and college learning

1.22 Medical Simulators

Medical simulators provide a **safe and effective way** to train medical students by replacing **real patients** with **models or computer-based simulations**.

Through simulation, students can practice repeatedly, make mistakes safely, and learn procedures **more effectively and confidently**.

Examples

1. Blood drawing practice
2. Laparoscopic surgery simulation
3. Trauma care training

Benefits of Medical Simulation

- Completely **safe** for patients
- Training focuses only on **essential skills**
- Builds **confidence** in students
- Supports **iterative learning** – assess, correct, and improve repeatedly

Types of Medical Simulators

1. **Plastic life-size models**
 - Respond to drugs
 - Simulate emergency conditions
2. **Computer graphic simulators**
 - Visualize medical procedures
 - Interactive and responsive
3. **Earlier models**

- Made of clay or stone
- Used mainly to demonstrate anatomy

4. **Active models**

- Reproduce living anatomy

5. **Interactive models**

- Respond directly to student decisions and actions

Key exam point:Simulation provides **safe, cost-effective, and repeatable training**, and is extensively used in **military, medical, space, and management education**.